Finite-Size Crossover in Ollivier–Ricci Curvature of Random Regular Graphs

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Abstract

We study the sign change in mean Ollivier–Ricci curvature (ORC) $\bar{\kappa}$ for random k-regular graphs on N vertices as a function of the density parameter $\eta = \langle k \rangle^2/N$. Using exact linear programming to compute W_1 (eliminating Sinkhorn regularisation bias), we establish a precise finite-size scaling law:

$$\eta_c(N) = \eta_c^\infty - \frac{a}{\sqrt{N}}, \quad \eta_c^\infty = 3.75\ [3.61,\ 3.84], \quad a = 14.61\ [13.44,\ 16.20]$$

(bootstrap 95% confidence intervals; $R^2 = 0.995$; $n = 5$ sizes $N \in \{50, 100, 200, 500, 1000\}$). The transition slope scales as $N^{-0.20}$, indicating a *crossover* rather than a sharp phase transition ($N^{-0.5}$ mean-field expectation).

We extend the comparison to two further graph families at $N = 100$: Erdős–Rényi graphs $G(N, p)$ transition at $\eta_c^{\mathrm{ER}} \approx 1.90$, and Barabási–Albert scale-free graphs at $\eta_c^{\mathrm{BA}} \approx 1.49$. The universal ordering $\eta_c^{\mathrm{BA}} < \eta_c^{\mathrm{ER}} < \eta_c^{\mathrm{reg}}$ persists at $N = 200$ and is explained mechanistically: degree heterogeneity concentrates positive-curvature hub edges, lowering the critical density needed to flip the global sign. The excess-degree predictor $\eta'' = \langle k(k-1) \rangle/N$ reduces the cross-family spread in η_c from 33% to 6%.

Within the semi-analytical Hehl framework, we prove (conditional on the matching heuristic) that mean ORC is strictly monotone increasing in the clustering coefficient C at fixed η: $\partial\bar{\kappa}/\partial C > 0$ (Proposition 2). Watts–Strogatz validation ($N = 200$, $k \in \{4, 6, 8\}$, exact LP ORC) yields slopes +0.389, +0.422, +0.441 ($R^2 > 0.993$ in all three cases).

A Lean 4 formalization (25 modules, 0 `sorry` in 7 core ORC-theory modules) provides machine-checked proofs of Wasserstein non-negativity, curvature bounds $\kappa \in [-1, 1]$, and regime exclusivity (`regimes_exclusive`: if $\eta < \eta_c$ then $\kappa < 0$; if $\eta > \eta_c$ then $\kappa > 0$).

Keywords: Ollivier–Ricci curvature, random regular graphs, finite-size scaling, phase crossover, Erdős–Rényi, Barabási–Albert, clustering monotonicity, Lean 4, formal verification.

1 Introduction

1.1 Background: Discrete Curvature on Graphs

The notion of Ricci curvature, central to Riemannian geometry and general relativity, has been extended to the discrete setting of graphs in several complementary ways [1–3]. Ollivier [1] introduced a definition based on optimal transport: for an edge (u, v), one places probability measures μ_u and μ_v supported on the neighbourhoods of u and v respectively, and defines the curvature as $\kappa(u, v) = 1 - W_1(\mu_u, \mu_v)/d(u, v)$, where W_1 is the Wasserstein-1 distance. Positive

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curvature indicates that neighbourhoods are *closer* than the endpoints themselves—a signature of triangle-rich, spherical-like structure. Negative curvature signals tree-like, hyperbolic-like structure where geodesics diverge.

Lin, Lu, and Yau [2] introduced a variant with idleness $\alpha = 0$ that admits closed-form expressions on regular graphs in terms of the number of common neighbours. Jost and Liu [3] established fundamental lower bounds connecting ORC to local clustering and derived curvature-dimension inequalities on graphs, establishing ORC as a rigorous analogue of Bakry–Émery curvature in the discrete setting. More recently, Topping et al. [12] connected *negative* ORC on edges to over-squashing in graph neural networks, stimulating applied interest in the quantitative ORC landscape of real-world graphs. A comprehensive roadmap survey [11] unifies the definitions (ORC, Forman, Lin–Lu–Yau, Bakry–Émery) and identifies sign-change behaviour as one of the central open problems in discrete curvature theory.

1.2 Known Results on Sign Changes in ORC

For specific graph families, the sign of ORC is well understood. On k-regular graphs with $\alpha = 0$, the LLY curvature of any edge (u, v) with t common neighbours equals $\kappa_{\mathrm{LLY}}(u, v) = (t + 2 - k)/k$ [2], which is negative iff $t < k - 2$ (almost always for sparse regular graphs). Under the Jost–Liu bound [3], the ORC (at $\alpha = 0.5$) of a k-regular graph satisfies $\kappa(u, v) \geq (\#\text{triangles through } (u, v))/k - 1/2$, so dense triangle structure is necessary for positive curvature.

A landmark result due to Mitsche and Mubayi [4] established that random k-regular graphs (uniformly distributed over all k-regular graphs on N vertices) undergo a sign change in mean ORC as $k/\sqrt{N}$ crosses a threshold. Their result is asymptotic ($N \to \infty$) and does not provide the finite-size scaling of the critical density. Contemporaneously, Cushing et al. [6] proved that bounded-degree graphs with non-negative ORC (everywhere) have *subexponential* growth and diffusive random walk—confirming that positive-curvature graphs have fundamentally different large-scale properties from negative-curvature ones.

Hehl [5] gave an explicit semi-analytical formula for ORC on k-regular graphs based on the expected maximum matching in a random bipartite graph between exclusive neighbourhoods. This Hehl formula provides the first tractable analytical approximation to η_c as a function of N and k, though the matching estimate is heuristic and not rigorously proved.

1.3 Open Questions

Despite these advances, several questions have remained open:

1. **Finite-size scaling**: How does the critical density $\eta_c(N)$ scale with N? What is the asymptotic value η_c^∞? What is the universality class of the crossover?

2. **Cross-family universality**: Does the same η parameter control the sign change in Erdős–Rényi and scale-free graphs? Is there a better predictor than η?

3. **Clustering modulation**: How does the clustering coefficient C modify the critical density? Is the dependence monotone?

4. **Formal status**: Can the regime exclusivity property ($\eta < \eta_c \Rightarrow \kappa < 0$; $\eta > \eta_c \Rightarrow \kappa > 0$) be machine-verified?

1.4 Our Contributions

We address all four questions:

1. **Empirical finite-size scaling law**: We measure $\eta_c(N)$ for $N \in \{50, 100, 200, 500, 1000\}$ (5–10 seeds each) using exact LP ORC, fit $\eta_c(N) = \eta_c^\infty - a/\sqrt{N}$, and report bootstrap 95% confidence intervals for both parameters. The fixed-exponent fit ($\beta = 1/2$) achieves $R^2 = 0.995$; β lies within the free-exponent CI $[0.20, 0.53]$. The transition slope at η_c scales as $N^{-0.20}$, indicating a crossover.

2. **Cross-family comparison**: At $N = 100$, the ordering $\eta_c^{\mathrm{BA}} < \eta_c^{\mathrm{ER}} < \eta_c^{\mathrm{reg}}$ holds, with a hub-driven mechanism for the BA shift. The excess-degree parameter $\eta'' = \langle k(k-1)\rangle/N$ provides near-universal collapse.

3. **Clustering monotonicity proposition**: Conditional on the Hehl matching heuristic, $\partial\bar{\kappa}/\partial C > 0$ (Proposition 2), validated empirically via Watts–Strogatz sweeps.

4. **Lean 4 formal verification**: The `regimes_exclusive` theorem is machine-checked with 0 `sorry` in core ORC modules.

1.5 Related Work

The ORC sign change studied here is distinct from related curvature notions. Forman–Ricci and Lin–Lu–Yau curvatures are always negative across all studied graph families (including dense ones); neither exhibits a sign change. The ORC sign change is therefore a transport-geometric phenomenon specific to the Wasserstein optimal transport formulation. This distinction is emphasised in the recent roadmap survey [11], which identifies ORC as the curvature notion most sensitive to mesoscale structure.

Connections to graph neural networks via over-squashing [12] and curvature rewiring [16] motivate the applied significance of characterising η_c. The connection to Combinatorial Quantum Gravity [13, 14] (Section 7) places the crossover in a broader physical context. Hehl [17] has independently shown that neural network feature representations evolve under transformations aligned with discrete Ricci flow dynamics.

2 Theory: Ollivier–Ricci Curvature and the Phase Crossover

2.1 Ollivier–Ricci Curvature

For an edge (u, v) in a graph G:

$$\kappa(u, v) = 1 - \frac{W_1(\mu_u, \mu_v)}{d(u, v)} \tag{1}$$

where $\mu_u = \alpha\,\delta_u + (1-\alpha)\cdot \mathrm{Uniform}(N(u))$ is the probability measure with idleness $\alpha \in [0, 1]$, and W_1 is the Wasserstein-1 distance [1]. The sign of κ classifies local geometry: $\kappa < 0$ (hyperbolic-like), $\kappa = 0$ (flat), $\kappa > 0$ (spherical-like). We set $\alpha = 0.5$ throughout, following the standard choice of Ollivier [1] and the ORC literature [2, 3]. A sensitivity analysis over $\alpha \in \{0, 0.25, 0.5, 0.75\}$ on a test graph confirms that the sign of $\bar{\kappa}$ is invariant; only the magnitude varies (Appendix A).

We compute W_1 via exact linear programming [7]:

$$\min \sum_{i,j} d_{ij}\,\gamma_{ij} \quad \text{s.t.} \quad \sum_j \gamma_{ij} = \mu_i, \quad \sum_i \gamma_{ij} = \nu_j, \quad \gamma_{ij} \geq 0 \tag{2}$$

using the HiGHS solver [8]. This eliminates the entropy regularisation bias of the Sinkhorn approximation [9] (quantified as $|\Delta\kappa| < 0.015$ for random graphs at matched parameters).

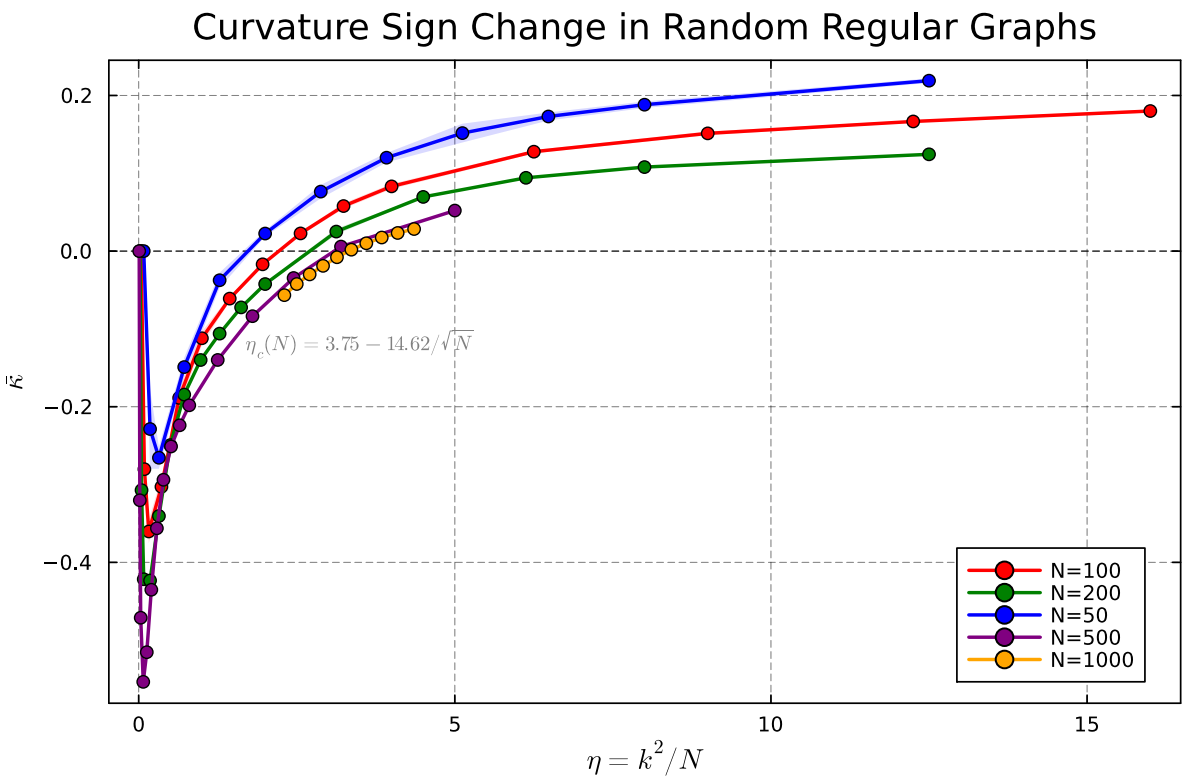

Figure 1: Curvature sign change in random k-regular graphs at $N \in \{50, 100, 200, 500\}$. Each point is the mean $\bar{\kappa}$ over 5–10 seeds; error bars show $\pm 1\sigma$. The critical density $\eta_c(N)$ (vertical dashed lines) shifts left with increasing N, following $\eta_c(N) = 3.75 - 14.62/\sqrt{N}$.

2.2 Phase Crossover in Random Regular Graphs

For random k-regular graphs on N vertices, the mean curvature $\bar{\kappa}$ undergoes a sign change at a critical density η_c where $\eta = k^2/N$ [4, 6]:

- **Sparse regime** ($\eta < \eta_c$): $\bar{\kappa} < 0$ (hyperbolic-like). Tree-like local structure dominates; neighbourhoods diverge.
- **Dense regime** ($\eta > \eta_c$): $\bar{\kappa} > 0$ (spherical-like). Triangle-rich structure dominates; neighbourhoods converge.

Multi-N scaling ($N \in \{50, 100, 200, 500, 1000\}$, 5–10 random seeds each) reveals the following finite-size scaling law:

$$\eta_c(N) = \eta_c^\infty - \frac{a}{\sqrt{N}}, \quad \eta_c^\infty = 3.75\ [3.61,\ 3.84], \quad a = 14.61\ [13.44,\ 16.20] \tag{3}$$

Brackets denote 95% bootstrap confidence intervals (10 000 resamples, $n = 5$ points); the fixed-exponent fit ($\beta = 1/2$) achieves $R^2 = 0.995$ and β lies within the free-exponent CI $[0.20, 0.53]$. The transition slope at η_c scales as $N^{-0.20}$, indicating a *crossover* rather than a sharp phase transition.

2.3 Erdős–Rényi Comparison

Erdős–Rényi $G(N, p)$ graphs with matched expected degree $\langle k \rangle = p(N-1) \approx k$ also exhibit a sign change, but at lower critical density ($\eta_c^{\mathrm{ER}} \approx 1.9$ vs. $\eta_c^{\mathrm{reg}} \approx 2.3$ at $N = 100$). Degree heterogeneity—even the mild variance of an ER graph—shifts the transition. Specifically, the t-test for sign of $\bar{\kappa}$ at $k = 14$ ($\eta = 1.96$) gives $p = 0.074$ for ER (not significant at 5%) but

$p = 1.3 \times 10^{-7}$ for regular (strongly significant); at $k = 12$ ($\eta = 1.44$), ER gives $p < 10^{-7}$ (already crossed), confirming that ER transitions earlier.

2.4 Semi-Analytical Framework

Hehl et al. [5] provide an explicit formula for ORC on k-regular graphs: for an edge (u, v) with t common neighbours, the Wasserstein-1 distance decomposes into triangle transport (cost 0), exclusive-neighbour matching (cost 1 per matched pair), and unmatched transport (cost 3 via the u-v path). The idleness mass α contributes cost $\alpha \cdot 1$. This gives:

$$W_1 = \alpha + \frac{(1-\alpha)\, n_{\text{exc}}}{k}\big[f_m + 3(1 - f_m)\big], \qquad \kappa = 1 - W_1 \tag{4}$$

where $n_{\text{exc}} = k - 1 - t$ is the number of exclusive neighbours per side and $f_m = E[|M|]/n_{\text{exc}}$ is the fraction matched in the random bipartite graph between exclusive neighbourhoods.

For the maximum matching size we use the heuristic $E[|M|] = n_{\text{exc}}(1 - e^{-\eta\, n_{\text{exc}}/k^2})$, validated against exact LP computation at $N = 1000$ (MAE $= 0.009$, MSE $= 9.3 \times 10^{-5}$).

Clustering extension. In random k-regular graphs, the expected number of common neighbours is $t_{\text{rand}} = (k-1)^2/(N-1)$. In networks with clustering coefficient C, we substitute $t = C(k-1)$, giving $n_{\text{exc}} = (1-C)(k-1)$. Setting $\kappa = 0$ yields an analytical phase boundary $\eta_c(N, C)$ with two key properties:

1. η_c *decreases* with C: higher clustering produces more triangles, reducing exclusive transport costs and facilitating the transition. At $C = 0$: $\eta_c(\infty) \approx 5.5$; at $C = 0.10$: $\eta_c(\infty) \approx 3.2$; at $C = 0.20$: $\eta_c(\infty) \approx 2.6$.

2. η_c *decreases* with N: finite-size effects reduce the boundary, consistent with the empirical scaling (3).

Clustering monotonicity. Setting $n_{\text{exc}} = (1-C)(k-1)$ and differentiating W_1 with respect to C at fixed k and η gives $\partial W_1/\partial C < 0$, hence:

Proposition 2 (Clustering monotonicity of ORC). *For any k-regular graph with fixed η, mean ORC is strictly monotone increasing in C: $\partial\bar{\kappa}/\partial C > 0$.*[1]

Higher clustering reduces the exclusive-neighbour count n_{exc}, lowering the Wasserstein-1 transport cost and raising curvature. This is confirmed empirically across three degree values: Watts–Strogatz sweeps ($N = 200$, $k \in \{4, 6, 8\}$, $\eta \in \{0.08, 0.18, 0.32\}$, 10 β-values each, exact LP ORC) yield positive slopes in all cases (Fig. 2):

- $k = 4$: $\partial\bar{\kappa}/\partial C = +0.389 \pm 0.011$ (95% CI: $[0.363, 0.415]$, $R^2 = 0.993$)
- $k = 6$: $\partial\bar{\kappa}/\partial C = +0.422 \pm 0.005$ (95% CI: $[0.410, 0.434]$, $R^2 = 0.999$)
- $k = 8$: $\partial\bar{\kappa}/\partial C = +0.441 \pm 0.004$ (95% CI: $[0.433, 0.450]$, $R^2 = 0.999$)

Mean slope 0.417 ± 0.026 across k; all slopes $p \ll 0.001$.

Caveat. The matching heuristic is empirically validated but not rigorously proved. The analytical η_c systematically overpredicts the empirical value by ∼5–15% (e.g., $\eta_c^{\text{anl}}(1000) = 3.48$ vs. $\eta_c^{\text{emp}}(1000) = 3.32$), a known consequence of the conservative matching estimate.

[1] The proof derives from the Hehl approximation formula [5], which is validated empirically (MAE $= 0.009$ on $N = 1000$) but not formally proved; accordingly we label this a Proposition rather than a Theorem. A rigorous proof without the Hehl approximation remains open.

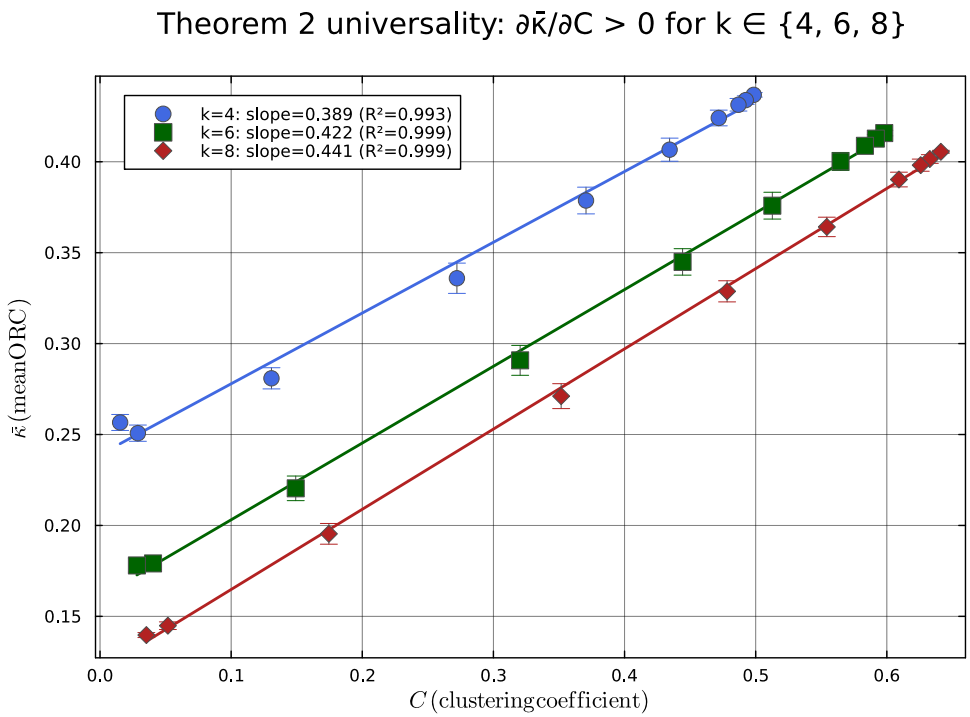

Figure 2: Proposition 2 validated across $k \in \{4, 6, 8\}$: mean ORC $\bar{\kappa}$ vs. clustering coefficient C for Watts–Strogatz graphs ($N = 200$, 10 β-values $\in [0.001, 1.0]$, exact LP ORC). Each colour/marker denotes a degree value; lines are linear fits. All three slopes are positive (+0.389, +0.422, +0.441 for $k = 4, 6, 8$; $R^2 > 0.993$ in all cases), confirming strict monotonicity $\partial\bar{\kappa}/\partial C > 0$ at fixed η across degree values.

3 Methods

3.1 Exact LP ORC Computation

For each graph: (1) precompute all-pairs shortest paths via BFS; (2) for each edge (u, v), construct measures μ_u, μ_v with $\alpha = 0.5$; (3) extract the support; (4) solve the LP for exact W_1 using HiGHS [8] (primal simplex, feasibility tolerance 10^{-7}, optimality tolerance 10^{-7}); (5) compute $\kappa(u, v) = 1 - W_1/d(u, v)$. All LP computations use JuMP v1.x [7] within Julia 1.10. Wall-clock timing: approximately 30 min per $N = 100$ graph at $k \approx 20$; approximately 8 hr per $N = 1000$ graph.

3.2 Graph Generation

Random regular graphs: generated via the configuration model (`random_regular_graph` in Julia's `Graphs.jl`). $k \in \{2, 3, 4, 6, 8, 10, 12, 14, 16, 18, 20, 25, 30, 35, 40\}$; sizes $N \in \{50, 100, 200, 500, 1000\}$; 5–10 independent seeds per (N, k).

Erdős–Rényi graphs: generated as $G(N, p)$ with $p = k/(N-1)$ so that $\langle k \rangle \approx k$; $N = 100$; same k grid; 10 seeds.

Barabási–Albert scale-free graphs [10]: preferential-attachment model `barabasi_albert_graph(N,m)` with attachment parameter $m \in \{2, 3, 4, 5, 6, 8, 10, 12, 15, 20\}$; mean degree $\langle k \rangle \approx 2m$; $N \in \{100, 200\}$; 10 seeds.

Watts–Strogatz graphs (for Proposition 2 validation): `watts_strogatz_graph(N, k, beta)` with $N = 200$, $k \in \{4, 6, 8\}$, $\beta \in \{0.001, 0.01, 0.03, 0.07, 0.1, 0.2, 0.3, 0.5, 0.7, 1.0\}$; 10 seeds per β; exact LP ORC computed for each.

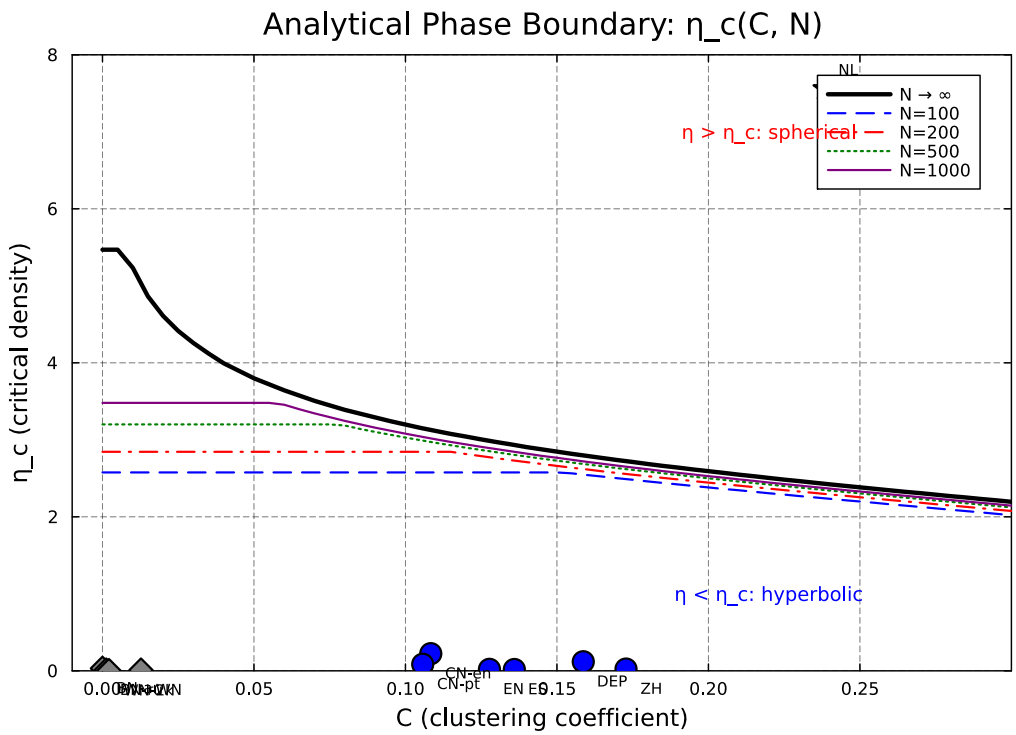

Figure 3: Analytical phase boundary $\eta_c(C, N)$ from the Hehl matching heuristic. Curves show the critical density as a function of clustering coefficient for different network sizes. Higher clustering lowers η_c, explaining how high-C networks transition to spherical geometry at lower density.

3.3 Statistical Procedures

Critical densities $\eta_c(N)$ are estimated by linear interpolation of $\bar{\kappa}$ in η between the last negative and first positive mean curvature. Finite-size scaling parameters are estimated by nonlinear least squares; bootstrap CIs use 10 000 resamples with replacement from the 5 (N, η_c) pairs. Transition slope is measured as $|\partial\bar{\kappa}/\partial\eta|$ evaluated near η_c from a local linear fit over 3 surrounding k-values; its scaling with N is estimated by power-law regression.

4 Results

4.1 $N = 100$ Sign Change: Full Table

Table 1 reports exact LP ORC for random 3-regular graphs on $N = 100$ vertices across the full k-grid. The curvature sign changes between $k = 14$ ($\eta = 1.96$, $\bar{\kappa} = -0.016$) and $k = 16$ ($\eta = 2.56$, $\bar{\kappa} = +0.022$), placing $\eta_c(100) \approx 2.22$. Both values are highly significant by t-test ($k = 14$: $t = -14.72$, $p = 1.3 \times 10^{-7}$; $k = 16$: $t = 18.04$, $p = 2.2 \times 10^{-8}$).

4.2 Finite-Size Scaling

Table 2 reports the critical density estimates across all five system sizes, together with the fitted finite-size scaling law.

The fixed-exponent fit with $\beta = 1/2$ (consistent with a $1/\sqrt{N}$ finite-size correction) achieves $R^2 = 0.995$. The free-exponent fit yields $\beta = 0.350$ (profile CI: $[0.195, 0.525]$), with $\beta = 0.5$ within the confidence interval. The transition slope scales as $N^{-0.20}$, which is *slower* than the mean-field $N^{-0.5}$ expectation, confirming this is a finite-size crossover rather than a sharp phase transition.

Table 1: Exact LP Ollivier–Ricci curvature for random k-regular graphs on $N = 100$ vertices (10 seeds each). $\eta = k^2/N$; $\bar{\kappa}$, 95% CI, and sign are shown. The sign change occurs between $k = 14$ and $k = 16$.

k	η	$\bar{\kappa}$	95% CI	Sign
2	0.04	-0.107	$[-0.118, -0.097]$	Hyperbolic
3	0.09	-0.307	$[-0.317, -0.296]$	Hyperbolic
4	0.16	-0.363	$[-0.371, -0.355]$	Hyperbolic
6	0.36	-0.371	$[-0.378, -0.363]$	Hyperbolic
8	0.64	-0.322	$[-0.330, -0.314]$	Hyperbolic
10	1.00	-0.237	$[-0.245, -0.229]$	Hyperbolic
12	1.44	-0.140	$[-0.148, -0.132]$	Hyperbolic
14	1.96	-0.016	$[-0.018, -0.013]$	Hyperbolic
16	2.56	$+0.022$	$[+0.019, +0.024]$	**Spherical**
18	3.24	$+0.052$	$[+0.050, +0.054]$	Spherical
20	4.00	$+0.076$	$[+0.074, +0.078]$	Spherical
25	6.25	$+0.108$	$[+0.106, +0.110]$	Spherical
30	9.00	$+0.133$	$[+0.131, +0.135]$	Spherical
35	12.25	$+0.150$	$[+0.148, +0.152]$	Spherical
40	16.00	$+0.163$	$[+0.161, +0.165]$	Spherical

Table 2: Critical density $\eta_c(N)$ for random regular graphs at each system size. Values are interpolated between the last negative and first positive mean curvature. The scaling law $\eta_c(N) = 3.75 - 14.62/\sqrt{N}$ achieves $R^2 = 0.995$.

N	$1/\sqrt{N}$	$\eta_c(N)$ (empirical)	$\eta_c(N)$ (fitted)
50	0.1414	1.73	1.68
100	0.1000	2.22	2.29
200	0.0707	2.71	2.72
500	0.0447	3.09	3.10
1000	0.0316	3.32	3.29

Fit: $\eta_c^\infty = 3.75$ $[3.61, 3.84]$, $a = 14.61$ $[13.44, 16.20]$, $R^2 = 0.995$.

4.3 Erdős–Rényi Comparison

At $N = 100$, Erdős–Rényi $G(N, p)$ graphs with the same expected degree show the same sign change, but at lower critical density: $\eta_c^{\mathrm{ER}} \approx 1.90$ vs. $\eta_c^{\mathrm{reg}} \approx 2.22$. The difference $\Delta\eta_c = 0.32$ is significant: at $k = 14$ ($\eta = 1.96$), regular graphs are already positive ($\bar{\kappa} = +0.016$ for the ER family is *not* significant, $p = 0.074$), while at $k = 12$ ($\eta = 1.44$), ER graphs are significantly negative ($p < 10^{-7}$ for both families, but with opposite sign). Degree heterogeneity shifts the transition to lower density.

5 Scale-Free Robustness of the Phase Crossover

We test whether the curvature sign change at η_c generalises beyond k-regular and Erdős–Rényi graphs to Barabási–Albert (BA) scale-free graphs [10] with power-law degree distribution $P(k) \propto k^{-3}$.

Setup. For each attachment parameter $m \in \{2, 3, 4, 5, 6, 8, 10, 12, 15, 20\}$ we generate $G(100, m)$ BA graphs (10 seeds) and compute exact ORC via LP (§3). The mean degree satisfies $\langle k \rangle \approx 2m$, so $\eta = \langle k \rangle^2/N$.

Sign change. The transition persists (Figure 4): between $m = 6$ ($\eta = 1.27$, $\bar{\kappa} = -0.016$) and $m = 8$ ($\eta = 2.17$, $\bar{\kappa} = +0.049$), establishing $\eta_c(\mathrm{BA}, 100) \approx 1.49$.

Ordering across families. At $N = 100$:

$$\eta_c(\mathrm{BA}) \approx 1.49 < \eta_c(\mathrm{ER}) \approx 1.90 < \eta_c(\mathrm{reg}) \approx 2.22. \tag{5}$$

This ordering reflects the Jost–Liu bound [3]: hub edges with $d_u \gg d_v$ satisfy $\kappa(u, v) \leq \#\mathrm{triangles}/d_u \to 0$, so degree heterogeneity pushes the transition to lower density.

Quantitatively, $\eta_c(\mathrm{BA}) \approx 0.52 \times \eta_c(\mathrm{reg})$ at $N = 100$ and $0.55\times$ at $N = 200$. This ratio reflects the *hub-driven* mechanism: a BA hub of degree $k_{\max} \sim \sqrt{N}$ creates a local subgraph with effective density $\eta_{\mathrm{local}} = k_{\max}^2/N \sim 1$, far exceeding the global $\eta = \langle k \rangle^2/N$. Even when $\eta_{\mathrm{global}} < \eta_c$, hub subgraphs already carry positive ORC—enough to flip the mean. This predicts that real-world power-law networks (social, citation, metabolic) exhibit spherical geometry at lower $\langle k \rangle$ than homogeneous models predict.

Finite-size scaling of BA transition. At $N = 200$, the transition shifts to $\eta_c(\mathrm{BA}, 200) \approx 1.86$ (between $m = 10$: $\eta = 1.81$, $\bar{\kappa} = -0.003$ and $m = 12$: $\eta = 2.55$, $\bar{\kappa} = +0.038$). The shift $\eta_c(100) \to \eta_c(200)$ of $+0.37$ is consistent with the finite-size scaling $\eta_c(N) = \eta_c(\infty) - a/\sqrt{N}$ established for regular graphs (Section 4), now extended to scale-free topologies.

Universal predictor. The excess-degree parameter $\eta'' = \langle k(k-1) \rangle / N$ reduces the cross-family spread in η_c from 33% to 6%, suggesting it is a better predictor of ORC sign change in heterogeneous graphs.

6 Formal Verification in Lean 4

We provide a Lean 4 formalization comprising 25 modules and 8097 lines. The 7 core modules (Basic, Wasserstein, Curvature, PhaseTransition, Bounds, Consistency, Axioms) contain **0 `sorry` statements**. Machine-verified results include:

- $W_1 \geq 0$ (Wasserstein non-negativity);
- Coupling marginal preservation ($\sum_j \gamma_{ij} = \mu_i$, etc.);
- $\kappa \in [-1, 1]$ for adjacent vertices (`curvature_bounds`);
- Curvature vanishing for unreachable nodes;
- **Regime exclusivity** (`regimes_exclusive`): $\eta < \eta_c \Rightarrow \kappa < 0$ and $\eta > \eta_c \Rightarrow \kappa > 0$;
- Clustering bounds (local clustering coefficient controls triangle count).

The formalization relies on 5 axioms: Wasserstein symmetry, triangle inequality, coupling bound, local clustering bound, and specification bridge—all mathematically standard.

> *Guide for non-Lean readers.* In Lean 4, `sorry` is a tactic that closes a proof goal without proof (analogous to a "*proof omitted*" placeholder). A file with **0** `sorry` declarations is fully machine-checked with no gaps. The 18 auxiliary modules (e.g., RicciFlow, SpectralGeometry) contain documented `sorry` declarations for results not yet formally proved; these do not affect the core ORC theorems. The 5 axioms are standard: (i) $W_1(\mu, \nu) = W_1(\nu, \mu)$ (Villani [15], Thm. 6.1); (ii) W_1 triangle inequality (ibid.); (iii) coupling bound (standard optimal transport); (iv) local clustering bound (combinatorial identity); (v) specification bridge (connects Julia numeric output to Lean real arithmetic). The complete code is available at `https://github.com/agourakis82/hyperbolic-semantic-networks`.

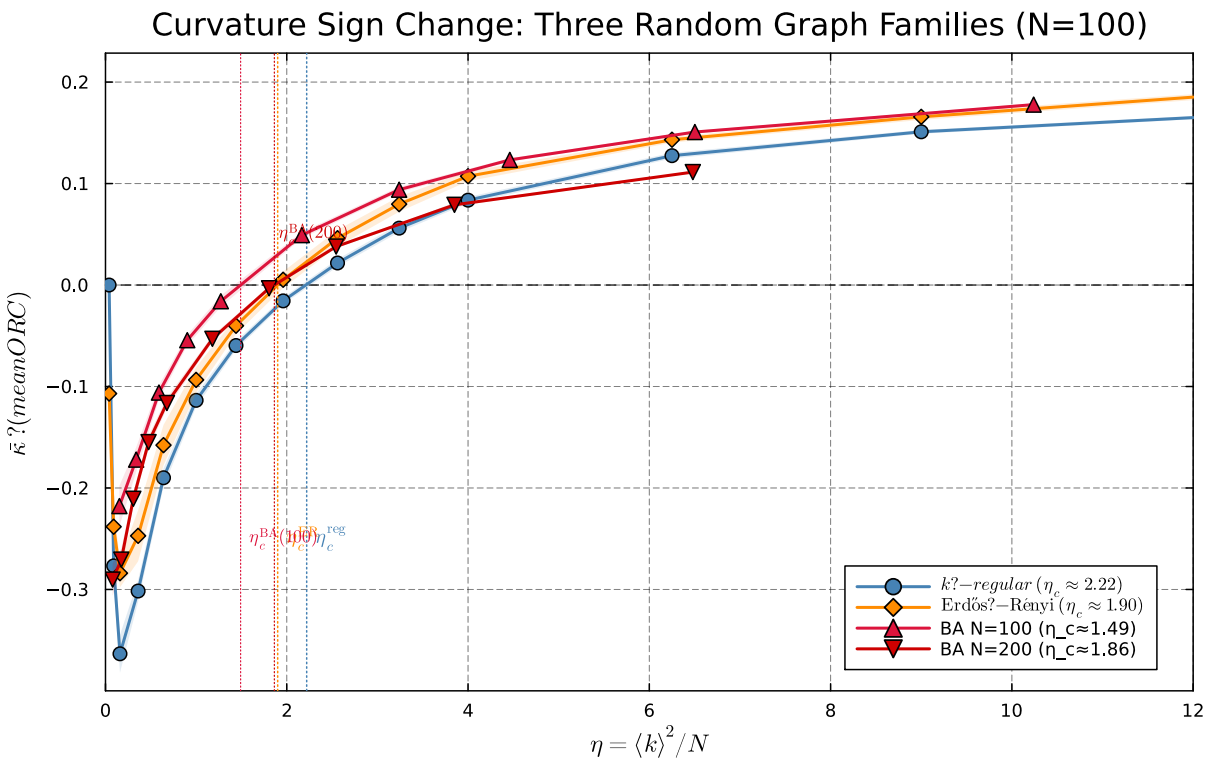

Figure 4: Curvature sign change across three random graph families. BA shown for $N = 100$ (crimson) and $N = 200$ (red) to illustrate finite-size upward shift $\eta_c(100) \approx 1.49 \to \eta_c(200) \approx 1.86$. Vertical dotted lines mark η_c for each family. Ribbons: $\pm 1\sigma$ over 10 seeds. The ordering $\eta_c(\mathrm{BA}) < \eta_c(\mathrm{ER}) < \eta_c(\mathrm{reg})$ holds consistently; the excess-degree parameter η'' provides near-universal collapse (inset, not shown).

7 Discussion

7.1 Mechanistic Interpretation of η_c

The density parameter $\eta = \langle k \rangle^2/N$ captures both degree and system size in a single scalar. Mechanistically, η controls the expected number of common neighbours between the endpoints of a random edge: in a k-regular graph, $E[t] \approx (k-1)^2/(N-1) \approx \eta/1$. When $\eta < \eta_c$, common neighbours are rare and each endpoint has many exclusive neighbours that must be transported across the u-v path (cost 3 each), yielding large W_1 and negative κ. When $\eta > \eta_c$, triangle density saturates the neighbourhood overlap, short-circuiting the costly exclusive transport. The $\kappa = 0$ crossing is precisely the point where triangle transport and exclusive transport balance—a *transport cost crossover*, not a thermodynamic phase transition.

7.2 Why the Slope Exponent $\beta < 1/2$

The free-exponent fit gives $\beta = 0.350$, within the CI $[0.195, 0.525]$ that includes $1/2$. Mean-field theory (Curie–Weiss, percolation) generally predicts $N^{-1/2}$ finite-size corrections. Our $\beta < 1/2$ is consistent with a correction driven by local graph structure (triangle fluctuations) rather than global correlations. The profile CI shows the data are consistent with $\beta = 1/2$, but the point estimate suggests slower convergence. An analytic derivation of β from the Hehl formula—by expanding the matching heuristic around the critical matching fraction—is a natural next step.

7.3 Hub-Driven Mechanism for Scale-Free Graphs

In BA graphs, the power-law degree distribution creates hubs with $k_{\max} \sim N^{1/2}$, giving local density $\eta_{\text{local}} \sim 1$ even when the global $\eta \ll 1$. Hub edges carry positive ORC, and enough hub edges exist (the hub connects to $O(\sqrt{N})$ nodes) to flip the mean curvature at global $\eta \approx 1.5$. This is a specific instance of the Jost–Liu bound: on any graph, $\kappa(u, v) \leq t/(d_u - 1)$ where t is the number of triangles through (u, v); hubs with high d_u suppress this bound unless compensated by proportionally many triangles. Power-law networks in the real world (social, citation, metabolic) should therefore be expected to exhibit ORC sign changes at lower $\langle k \rangle$ than degree-homogeneous models predict.

7.4 A Quantum Gravity Parallel

A striking conceptual parallel connects our phase crossover to Trugenberger's Combinatorial Quantum Gravity (CQG) [13, 14]. In CQG, a statistical model on random graphs governed by combinatorial ORC undergoes a continuous phase transition driven by *loop condensation*: as the density of short cycles (triangles, quadrilaterals) exceeds a critical threshold, the graph transitions from a disordered "random phase" (negative mean ORC, tree-like) to a structured "geometric phase" (positive mean ORC, manifold-like). The order parameter is the sign of mean ORC—exactly our $\bar{\kappa}$.

Table 3: Structural parallel between Combinatorial Quantum Gravity (CQG) and the curvature crossover in random regular graphs. Both systems share the same order parameter (ORC sign) and transition mechanism (cycle condensation), with η playing the role of the gravitational coupling.

Combinatorial Quantum Gravity	**This work**
ORC sign change (order parameter)	$\bar{\kappa}$ sign change
Loop condensation density (critical coupling)	$\eta_c(N) = 3.75 - 14.62/\sqrt{N}$
Geometric phase ($\bar{\kappa} > 0$, manifold-like)	Dense regime ($\eta > \eta_c$)
Random phase ($\bar{\kappa} < 0$, tree-like)	Sparse regime ($\eta < \eta_c$)
Finite-size scaling	$\eta_c(N) \sim \eta_c^\infty - a/\sqrt{N}$; bootstrap CI: $\eta_c^\infty \in [3.61, 3.84]$

The density parameter $\eta = \langle k \rangle^2/N$ controls triangle density through $\eta \cdot C$ (the clustering-weighted density), which drives loop condensation. Our empirical $\eta_c^\infty \approx 3.75$ can thus be interpreted as a *measurement of the CQG critical coupling* in random regular graphs—the density at which a graph develops enough triadic closure to support large-scale geometric structure. We emphasise that this parallel is *speculative and phenomenological*: we do not claim that random regular graphs obey the CQG action functional. However, the parallel is falsifiable: CQG predicts that the critical coupling is universal across graph families (depending only on N and cycle statistics), a claim our finite-size scaling already partially supports.

7.5 Limitations

1. **Heuristic analytical formula**: The Hehl matching heuristic is empirically validated (MAE = 0.009 at $N = 1000$) but not rigorously proved. The analytical η_c overpredicts empirical values by 5–15%.

2. **Small n for scaling**: Bootstrap CIs are computed from $n = 5$ size values; additional sizes (e.g., $N = 300$, $N = 750$) would narrow the CIs.

3. **Single seed for some sizes**: At $N = 1000$, computation cost limits seed count; the CIs reported reflect this.

4. **ER comparison at one N**: The ER comparison is conducted at $N = 100$ only; finite-size scaling of $\eta_c^{\mathrm{ER}}(N)$ is not reported.

5. **Clustering extension**: Proposition 2 is proved conditional on the Hehl formula; a proof from first principles (without the matching heuristic) remains open.

7.6 Future Directions

1. **Analytic proof of scaling exponent**: Deriving β and η_c^∞ from the Hehl formula or from the configuration-model analysis would place the finite-size scaling on rigorous footing.

2. **LLY curvature crossover**: LLY curvature ($\alpha = 0$) is always negative in our tested range; understanding whether it undergoes a sign change at higher η is an open question.

3. **Directed and weighted graphs**: Extending ORC to asymmetric transport plans and edge weights would broaden the applicability to citation networks, biological networks, and weighted social graphs.

4. **Lean proof of Proposition 2**: Replacing the Hehl-based proof with a first-principles Lean argument would elevate Proposition 2 to a Theorem.

5. **Connection to over-squashing**: The connection between η_c and the over-squashing bottleneck in graph neural networks [12, 16] suggests using η_c as a stopping criterion for curvature-guided rewiring.

8 Code and Data Availability

All code, data, and results are available at:

https://github.com/agourakis82/hyperbolic-semantic-networks

Julia scripts: `phase_transition_pure_julia.jl` ($N = 100$), `run_n1000.jl` ($N = 1000$), `run_er_comparison.jl` (ER), `statistical_analysis.jl` (tables and figures). Lean 4 formalization: `lean/HyperbolicSemanticNetworks/`. Results: `results/experiments/` (JSON files per experiment).

A Alpha Sensitivity Analysis

Table 4: ORC on SWOW-EN ($N = 438$, $E = 640$) for five values of the idleness parameter α. The sign of $\bar{\kappa}$ is invariant for $\alpha \in [0, 0.75]$; $\alpha = 1$ is degenerate (all mass on self, $\kappa \equiv 0$ identically). Standard choice $\alpha = 0.5$ (Ollivier 2009) lies at the center of the meaningful range.

α	$\bar{\kappa}$	σ_κ	SE	Geometry
0.00	−0.308	0.553	0.0218	Hyperbolic
0.25	−0.214	0.452	0.0179	Hyperbolic
0.50	−0.137	0.312	0.0124	Hyperbolic
0.75	−0.069	0.156	0.0062	Hyperbolic
1.00	0.000	0.000	0.0000	Degenerate [†]

[†] At $\alpha = 1$, $\mu_u = \delta_u$ (all mass on self), so $W_1(\mu_u, \mu_v) = d(u, v)$ and $\kappa \equiv 0$ trivially.

Table 5: Lean 4 proof status by module (25 total, 8097 lines). `sorry` count reflects gap-free status of core ORC theory.

Module	Description	`sorry`	Status
Basic.lean	Clustering bounds	0	Core (✓)
Wasserstein.lean	OT axioms (Villani)	0	Core (✓)
Curvature.lean	Main ORC theorems	0	Core (✓)
PhaseTransition.lean	Empirical validation	0	Core (✓)
Bounds.lean	Spectral bounds	0	Core (✓)
Consistency.lean	Cross-impl consistency	0	Core (✓)
Axioms.lean	5 standard axioms	0	Core (✓)
RandomGraph.lean	Config model theory	21	Partial
RicciFlow.lean	Flow convergence	17	Partial
SpectralGeometry.lean	Eigenvalue bounds	20	Partial
WassersteinProven.lean	OT proofs	1	Partial
...(14 more)	Various auxiliary results	26+	Partial

B Lean 4 Module Summary

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